

Fig. 2: An actual-size, single-side PCB for the signal diode based fire sensor

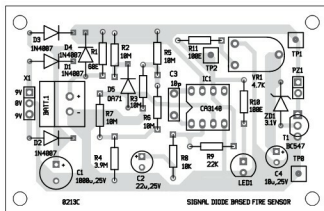


Fig. 3: Component layout for the PCB

The non-inverting input of IC1 is biased by a chain of high-value resistors R5, R6 and R7. Resistors R3 and R4 are connected from the resistor chain to ground to make the non-inverting input stable for an easy swing when the reverse current from the signal diode enters pin 3. Feedback resistor R9 sets the gain of the inverting amplifier. The inverting input is connected to ground through R8 and C2, which further increases the negative feedback. Capacitor C3 is connected between Strobe and Null pins for phase compensation. This arrangement makes the inverting amplifier highly sensitive, giving a high output for a small rise in the voltage at its non-inverting input. The output from the inverting amplifier drives transistor T1 to turn the alarm 'on' or activate a relay (not shown here).

Working of the circuit is simple. When diode D5 detects light or heat from fire, the output of IC1 gradually increases. To avoid false alarm, a zener diode (ZD1) based switch is

used. The zener diode conducts only when the output voltage from IC1 increases above 4 volts. That is, only a persisting and increasing fire will activate the buzzer. When the light from the fire increases, the output of IC1 also increases above 4 volts and the zener diode conducts. This triggers T1 and the buzzer sounds to indicate a fire hazard.

The alarm section can be further improved to operate a warning light, etc by using a relay at the collector of T1.

Construction and testing

An actual-size, single-side PCB for signal diode based fire sensor is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on

the PCB, enclose it in a suitable case. Fix the diode in the probable place of fire breakout and connect the diode to the circuit using thin wires.

More than one diode can be connected to the circuit in order to detect fire at different places. The only condition is that, since the diode detects infrared light, direct sunlight or light from an incandescent bulb should not fall on the diode. Light from fluorescent lamps will not trigger the alarm.

The buzzer will give a chirping sound if power supply has ripples. If battery is used, sound from the buzzer will be loud and continuous. Adjust VR1 to the left end to get the breakdown voltage for the zener diode.

Check 9V at test point TP1 with respect to TP0 to verify correct power supply to the circuit in the presence and absence of mains supply. Light a matchstick near the diode and verify change in the voltage at TP2. ●

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